

Loan Approval & Customer Risk Analysis

An End-to-End Financial Data Analytics & Risk Assessment Project

1. Project Metadata

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Domain: Banking & Financial Services / Credit Risk

Tools & Technologies: Python (Pandas), MySQL, Power BI (DAX), WPS Office

Dataset Scale: 52,000 Loan Applications

2. Project Objective

The objective of this project is to analyze 52,000 loan applications and identify the key financial and demographic factors associated with loan approval and rejection outcomes.

The analysis focuses on Credit Score, Age, Income, Employment Status, Existing Loans, and Outstanding Debt to understand approval patterns, customer segments, and potential risk indicators within the dataset.

The project also aims to convert the analysis into business insights and an interactive Power BI dashboard that can help users understand loan approval performance and customer risk patterns.

3. Business Problem Statement

Financial institutions need to balance loan approval opportunities with credit risk management.

Analyzing loan application data can help identify patterns in approval decisions and understand how different applicant characteristics are associated with loan outcomes.

This project addresses the following business questions:

Risk Pattern Identification

Which applicant characteristics, such as Credit Score, Age, Income, and Employment Status, are associated with higher or lower approval rates?

Portfolio Analysis

What is the overall volume of loan applications, approval rate, rejection rate, and average requested loan amount?

Customer Segmentation

Which applicant groups show higher approval rates based on credit score and age?

Financial Analysis

How do income, existing loans, and outstanding debt compare between different approval groups?

Process Insights

Which applicant characteristics show meaningful differences in approval outcomes within the dataset?

4. Analytical Pipeline

The project follows an end-to-end data analytics workflow:

1. Data Ingestion & Cleaning — Python/Pandas

Loaded and reviewed the raw dataset, handled missing values, checked duplicate records, corrected data types, and prepared the dataset for analysis.

2. Exploratory Data Analysis — Python/Pandas

Analyzed loan approval patterns across Credit Score, Age, Employment Status, Income, Existing Loans, Outstanding Debt, and other applicant attributes.

3. Database Analysis — MySQL

Imported the cleaned dataset into MySQL and performed 12 business-focused SQL questions using filtering, aggregation, grouping, subqueries, CTEs, and window functions.

4. Business Insights

Converted the analysis results into meaningful observations related to loan approval patterns and customer segments.

5. Interactive Visualization — Power BI

Created an interactive dashboard using DAX measures, KPI cards, slicers, and charts to present the major findings in a clear visual format.

6. Recommendations

Developed possible business recommendations based on the patterns observed in the dataset.

5. Dataset Architecture & Preprocessing

Dataset Overview

The dataset contains 52,000 loan application records covering demographic, financial, employment, credit, and loan-related information.

Key Dataset Fields

Field

Description

Applicant_ID

Unique identifier for each loan application

Age

Age of the applicant

Annual_Income

Applicant's annual income in ₹

Credit_Score

Applicant's credit score

Employment_Status

Employment category such as Employed, Self-Employed, or Unemployed

Loan_Amount_Requested

Amount of loan requested

Existing_Loans

Number of existing loans

Outstanding_Debt

Current outstanding debt amount

Loan_Approval_Status

Loan decision: 1 = Approved, 0 = Rejected

Data Preprocessing

Before performing the analysis, the dataset was reviewed and cleaned using Python and Pandas.

Missing Value Check

Missing values were identified and handled during preprocessing. The final cleaned dataset contained no remaining missing values.

Duplicate Check

Duplicate records were checked and handled during the cleaning process. The final dataset contained no duplicate records.

Data Type Check

Numeric fields, categorical fields, and financial values were reviewed and converted into appropriate data types for analysis.

Data Consistency

Categorical values and other fields were reviewed for consistency before performing further analysis. The cleaned dataset was then used for Pandas analysis, MySQL analysis, and Power BI visualization.

6. Database Migration & SQL Analysis

The cleaned dataset was imported into MySQL and stored as a relational table for SQL-based business analysis.

The SQL analysis included:

Filtering

Aggregation functions

GROUP BY

HAVING

Subqueries

Common Table Expressions (CTEs)

Window functions

DENSE_RANK()

A total of 12 business questions were developed to analyze loan approval patterns, customer segments, income, employment, existing loans, and outstanding debt.

7. Exploratory Data Analysis

7.1 Credit Score Analysis

Credit Score showed a strong association with loan approval outcomes in this dataset.

Low Tier (<600): 14.98% approval rate

Medium Tier (600–750): 85.37% approval rate

High Tier (>750): 84.80% approval rate

Applicants with a credit score below 600 had an 85.02% rejection rate, while applicants with scores of 600 or above had substantially higher approval rates.

Key Finding

Credit Score showed one of the strongest differences in approval rates among the variables analyzed.

7.2 Age Group Analysis

Approval rates varied across the different age groups:

18–30: 63.35% approval

31–50: 72.58% approval

Above 50: 15.14% approval

Applicants above 50 had an 84.86% rejection rate in this dataset.

Key Finding

Applicants above 50 showed a substantially lower approval rate compared with the other age groups.

7.3 Employment & Income Analysis

Approval rates across employment categories were relatively close:

Self-Employed: 64.93%

Employed: 64.11%

Unemployed: 63.15%

The approval rates differed only slightly across these categories.

The Pandas analysis also showed a difference in average annual income between approved and rejected applicants:

Approved: approximately ₹96,000

Rejected: approximately ₹61,000

Key Finding

Employment category alone showed limited differentiation in approval rates within this dataset.

7.4 Existing Loans & Outstanding Debt

Existing Loans

Approval rates remained relatively similar:

Existing Loans

Approval Rate

0

63.62%

1

64.72%

2

64.15%

Outstanding Debt

Average outstanding debt was also very similar:

Approved: ₹14,965.69

Rejected: ₹15,019.20

Key Finding

Existing loan count and average outstanding debt showed relatively small differences between approval groups in this dataset.

8. SQL Business Questions & Query Analysis

Business Questions

Q1

What is the total volume of loan applications, overall approval rate, and average loan amount requested?

Q2

What is the exact split between approved and rejected applications?

Q3

How does the approval rate vary across Low, Medium, and High Credit Score tiers?

Q4

What is the application volume, approval rate, and rejection rate for applicants with a Credit Score below 600?

Q5

How does the approval rate vary across different age groups?

Q6

What is the application volume and rejection rate for applicants above 50 years?

Q7

How many applicants have Credit Score ≥ 600 and Age ≤ 50 , and what is their approval rate?

Q8

What percentage of the total portfolio is represented by this high-approval segment?

Q9

What are the average income, requested loan amount, and approval rate across employment categories?

Q10

Who are the top 3 highest-income approved applicants within each employment category?

Q11

Does the number of existing loans have a noticeable relationship with loan approval rates?

Q12

How do average, minimum, and maximum outstanding debt values compare between approved and rejected applicants?

9. Key SQL Findings

Overall Portfolio Performance

Total Applications: 52,000

Approved: 33,366

Rejected: 18,634

Approval Rate: 64.17%

Rejection Rate: 35.83%

Average Loan Amount: ₹21,102.70

Credit Score Findings

Credit Tier

Applications

Approved

Approval Rate

Low (<600)

15,468

2,317

14.98%

Medium (600–750)

12,484

10,657

85.37%

High (>750)

24,048

20,392

84.80%

The Low Credit Score group had an 85.02% rejection rate.

High-Approval Segment

Applicants meeting both conditions:

Credit Score ≥ 600 AND Age ≤ 50

Applications: 36,532

Approved: 31,049

Approval Rate: 84.99%

Portfolio Share: 70.25%

This segment represents a large portion of the dataset and shows a high approval rate.

Income & Employment Findings

The highest-income approved applicants were identified within each employment category using the DENSE_RANK() window function.

Peak annual incomes were approximately ₹1.5 lakh across the employment groups.

This shows that employment category alone does not fully describe the income levels observed in the dataset.

10.Power BI Dashboard



Dashboard Title: loan approval & customer risk analysis

The Power BI dashboard was designed to provide an interactive view of loan approval performance and customer risk patterns.

KPI Cards

The dashboard includes:

Total Applications: 52K

Approved Loans: 33K

Rejected Loans: 19K

Average Loan Amount: ₹21.10K

Approval Rate: 64.17%

Rejection Rate: 35.83%

Dashboard Visuals

Loan Approval Overview

Donut chart showing the overall distribution between approved and rejected applications.

Approval Rate by Credit Score Tier

Column chart comparing approval rates across Low, Medium, and High credit score groups.

Approval Rate by Age Group

Column chart showing approval rate differences across age groups.

Approval Rate by Employment

Bar chart comparing approval rates across employment categories.

Approval Rate by Existing Loans

Area chart showing approval rates for applicants with 0, 1, and 2 existing loans.

Average Outstanding Debt by Approval Status

Column chart comparing average outstanding debt between approved and rejected applicants.

Interactive Slicers

The dashboard includes slicers for:

Credit Score Tier

Age Group

City/Town

Education

Marital Status

Employment Status

These allow users to explore different applicant segments interactively

11. Key Business Insights

1. Credit Score Shows a Strong Association with Approval

Applicants with a Credit Score below 600 had an 85.02% rejection rate, while Medium and High Credit Score groups had approval rates of approximately 85%.

2. Approval Rates Vary Significantly Across Age Groups

The 31–50 age group had a 72.58% approval rate, while applicants above 50 had a 15.14% approval rate.

This indicates a substantial difference in approval outcomes across age groups within the dataset.

3. High-Approval Segment Represents a Large Share of Applications

Applicants with Credit Score ≥ 600 and Age ≤ 50 represented:

70.25% of applications

84.99% approval rate

This makes it an important segment for further analysis.

4. Employment Status Shows Limited Difference

Approval rates were relatively close across:

Self-Employed — 64.93%

Employed — 64.11%

Unemployed — 63.15%

Therefore, employment category alone showed limited differentiation in this dataset.

5. Existing Loans and Outstanding Debt Show Similar Results Across Groups

Approval rates remained around 64% for applicants with 0, 1, or 2 existing loans.

Average outstanding debt was also similar between approved and rejected groups.

12. Strategic Business Recommendations

1. Streamlined Processing for the High-Approval Segment

Applicants with Credit Score ≥ 600 and Age ≤ 50 represent 70.25% of applications and have an 84.99% approval rate.

This segment could be considered for streamlined initial processing, subject to normal verification, risk assessment, and applicable lending procedures.

2. Additional Assessment for Low-Credit Applicants

Applicants with a Credit Score below 600 have an 85.02% rejection rate.

Instead of relying only on Credit Score, additional factors such as verified income, repayment capacity, existing debt obligations, and other relevant financial information could be considered.

3. Broader Financial Assessment for Applicants Above 50

Applicants above 50 show an 84.86% rejection rate in this dataset.

A broader assessment including income stability, repayment capacity, existing debt, and other relevant financial indicators could provide additional information beyond age alone.

4. Focus on Financial Indicators Beyond Employment Category

Since approval rates across employment categories are relatively close, financial indicators such as:

Verified income

Repayment capacity

Existing debt

Credit history

can be considered together rather than relying heavily on employment category alone.

13. Project Artifacts

The project contains the following files:

Loan Dataset.csv — Original loan application dataset containing 52,000 records.

cleaned_loan_approval.csv — Cleaned dataset prepared for analysis.

loan_approval_pandas.ipynb — Python/Pandas cleaning and analysis workflow.

loan_approval_sql_analysis.ipynb — SQL analysis covering 12 business questions.

14. Final Project Conclusion

loan_approval_dashboard.png — Final Power BI dashboard export.

This project developed an end-to-end loan approval analytics workflow using Python/Pandas, MySQL, and Power BI across 52,000 loan applications.

The analysis identified strong differences in approval rates across Credit Score and Age groups within the dataset. In comparison, Employment Status, Existing Loans, and Average Outstanding Debt showed relatively smaller differences between approval groups.

The Power BI dashboard converts these findings into an interactive business view, allowing users to explore approval performance, customer segments, and key risk patterns.

Overall, the project demonstrates the complete data analytics process:

Data Cleaning → Exploratory Analysis → SQL Business Analysis → Business Insights → Power BI Visualization → Recommendations

The recommendations in this project are based specifically on the patterns observed in the dataset and can serve as areas for further analysis and assessment.